

Intrinsic robust multiscale filtering

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Abstract

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1 Introduction

2 Intrinsic multiscale filtering

Suppose that observed data Y is composed of a signal $X = (X_t)$ and a noise ε_t :

$$Y_t = X_t + \varepsilon_t.$$

2.1 Local contrast fit

Fix a **contrast** function $\rho(\cdot)$, a **kernel** function $K(\cdot)$, and a **bandwidth** h . For every t , compute

$$\tilde{S}_t = \operatorname{arginf}_x \sum_u \rho(Y_u - x) K_h(t - u),$$

where $K_h(y) = K(y/h)$.

For $\rho(y) = y^2/2$, the value $\tilde{S}_t$ is the **local mean** of Y_u for $u \in [t - h, t + h]$:

$$\tilde{S}_t = \frac{1}{N_h(t)} \sum_u Y_u K_h(t - u), \quad N_h(t) \stackrel{\text{def}}{=} \sum_u K_h(t - u),$$

The choice $\rho(y) = |y|$ leads to the **local median**.

This step results in the first IMF $\tilde{S}_t^{(1)} = \tilde{S}_t$ and the residuals $Y_t^{(2)} = Y_t - \tilde{S}_t$.

2.2 Iterative procedure

With $Y_t^{(1)} \stackrel{\text{def}}{=} Y_t$, at every step $k \geq 1$, fix h_k and compute

$$\tilde{S}_t^{(k)} = \operatorname{arginf}_x \sum_u \rho(Y_u^{(k)} - x) K_{h_k}(t - u), \quad t \in [0, 1].$$

(Here $Y_t^{(k)} = Y_{1+t}^{(k)}$ for all t .) Compute the new residuals

$$Y_t^{(k+1)} = Y_t^{(k)} - \tilde{S}_t^{(k)}.$$

At every step k , decrease h_k by a fixed factor a (e.g. $a = \sqrt{2}$):

$$h_{k+1} = h_k / a.$$

Continue until $k \leq k_{\max}$. Alternatively, fix the smallest bandwidth $h_{\min}$ and stop if $h_{k+1} < h_{\min}$.

2.3 Parameter choice and output

Sequence of bandwidths h_k start with $h_1 \asymp 1$ (very big, e.g. 0.1). The starting bandwidth should ensure that each window $[t - h_1, t + h_1]$ contains about half of all observations. This ensures that for the first IMF, almost all noise is smoothed out, and only gross features remain.

Kernel We apply the [Epanechnikov kernel](#):

$$K(u) = \frac{3}{4}(1 - |u|)_+^2.$$

Bandwidth factor The bandwidth factor a can be any value between one and two. Our default choice is $a = \sqrt{2}$ or $a = 2$.

Output The procedure delivers a sequence of IMF's

$$\tilde{S}^{(1)}, \tilde{S}^{(2)}, \dots, \tilde{S}^{(k_{\max})}.$$

This [collection of the IMFs](#) $(\tilde{S}^{(k)})$ can be used for classification/identification.

2.4 True IMFs

The computed IMFs can be viewed as empirical/noisy versions of their population counterparts (true IMFs) defined with $X_t^{(1)} = X_t$ and the same sequence h_k by

$$S_t^{(k)} = \operatorname{arginf}_x \sum_u \mathbb{E} \rho(Y_u^{(k)} - x) K_{h_k}(t - u).$$

The question under study is the quality of recovery

$$\|\tilde{S}^{(k)} - S^{(k)}\|^2 = \sum_{t=1}^n |\tilde{S}_t^{(k)} - S_t^{(k)}|^2$$

or

$$\sup_t |\tilde{S}_t^{(k)} - S_t^{(k)}|.$$

2.5 Theoretical study for local linear smoothing

Fix $\rho(u) = u^2/2$ corresponding to [local mean](#) procedure.

2.5.1 Linear representation of the IMFs

Define $W^{(k)}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ as a linear mapping with the rows

$$W^{(k)} = \left(\frac{1}{N_k(t)} K_{h_k}(t - u) \right)_{t, u \in [0, 1]} \quad (2.1)$$

with

$$N_k(t) \stackrel{\text{def}}{=} \sum_u K_{h_k}(t - u).$$

Also, define

$$\begin{aligned} \mathcal{A}^{(k)} &\stackrel{\text{def}}{=} (\mathbb{I} - W^{(k)}) \circ \dots \circ (\mathbb{I} - W^{(1)}) \\ \mathcal{W}^{(k)} &\stackrel{\text{def}}{=} W^{(k)} \mathcal{A}^{(k-1)}. \end{aligned}$$

Lemma 2.1. *Each $\tilde{S}^{(k)} \in \mathbb{R}^n$ is a linear transform of the residuals $Y^{(k)} \stackrel{\text{def}}{=} \mathcal{A}^{(k-1)} Y$:*

$$\tilde{S}^{(k)} = W^{(k)} Y^{(k)} = \mathcal{W}^{(k)} \mathcal{A}^{(k-1)} Y = \mathcal{W}^{(k)} Y, \quad (2.2)$$

Proof. The identities $\tilde{S}^{(k)} = W^{(k)} Y^{(k)}$ and $Y^{(k+1)} = Y^{(k)} - \tilde{S}^{(k)}$ imply

$$Y^{(k+1)} = Y^{(k)} - \tilde{S}^{(k)} = (\mathbb{I} - W^{(k)}) Y^{(k)}.$$

By induction, for $k \geq 1$,

$$Y^{(k+1)} = \prod_{j=1}^k (\mathbb{I} - W^{(j)}) Y = \mathcal{A}^{(k)} Y.$$

Similarly,

$$\tilde{S}^{(k+1)} = W^{(k+1)} \prod_{j=1}^k (\mathbb{I} - W^{(j)}) Y = W^{(k+1)} \mathcal{A}^{(k)} Y = \mathcal{W}^{(k+1)} Y,$$

and the assertion follows. $\square$

We can treat $(W^{(k)})$'s as a family of multiscale operators in $\mathbb{R}^n$ mapping the data Y to a family of noisy IMFs $(\tilde{S}^{(k)})$.

Lemma 2.2. *For every $k \geq 1$, the eigenvalues $\lambda_j(W^{(k)})$ of the operator $W^{(k)}$ satisfy*

$$\lambda_j(W^{(k)}) \in (0, 1), \quad \|W^{(k)}\| = \lambda_{\max}(W^{(k)}) = 1.$$

Furthermore, $\mathcal{A}^{(k)}$ from (2.2) fulfills $\|\mathcal{A}^{(k)}\| \leq 1$.

Proof. Each matrix $W^{(k)}$ is stochastic in the sense that the elements of each row of $W^{(k)}$ are non-negative and sum to one. The result follows by Gershgorin theorem. $\square$

2.5.2 Linear decomposition of the IMFs

Let $X_t = \mathbb{E}Y_t$, $X^{(1)} = X$.

Lemma 2.3. *For every $k \geq 1$, the true IMF $S^{(k)}$ can be represented as a linear transform of X :*

$$S^{(k)} = \mathcal{W}^{(k)}X = W^{(k)} \circ (\mathbb{I} - W^{(k-1)}) \circ \dots \circ (\mathbb{I} - W^{(1)})X. \quad (2.3)$$

Let $\boldsymbol{\varepsilon} \stackrel{\text{def}}{=} X - S$ and $\mathbb{E}\boldsymbol{\varepsilon} = 0$. Then for $k \geq 1$,

$$\tilde{S}^{(k)} - S^{(k)} = W^{(k)}\mathcal{A}^{(k-1)}\boldsymbol{\varepsilon} = \mathcal{W}^{(k)}\boldsymbol{\varepsilon},$$

where $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^{(0)} \stackrel{\text{def}}{=} Y - X$ and

$$\begin{aligned} \boldsymbol{\varepsilon}^{(k)} &\stackrel{\text{def}}{=} (\mathbb{I} - W^{(k)})\boldsymbol{\varepsilon}^{(k-1)} \\ &= (\mathbb{I} - W^{(k)}) \circ (\mathbb{I} - W^{(k-1)}) \circ \dots \circ (\mathbb{I} - W^{(1)})\boldsymbol{\varepsilon} = \mathcal{A}^{(k)}\boldsymbol{\varepsilon}. \end{aligned} \quad (2.4)$$

Proof. We can use

$$S^{(k)} = W^{(k)}X^{(k)}, \quad X^{(k+1)} = X^{(k)} - S^{(k)}.$$

Statement (2.3) follows similarly to (2.2). $\square$

2.5.3 Noise propagation

Let $\boldsymbol{\varepsilon}^{(k)} = \mathcal{A}^{(k)}\boldsymbol{\varepsilon}$ be given by (2.4). Obviously, $\mathbb{E}\boldsymbol{\varepsilon}^{(k)} = 0$. Define $\Sigma^{(k)} = \text{Var}(\boldsymbol{\varepsilon}^{(k)})$.

Lemma 2.4. *It holds for $k \geq 1$*

$$\Sigma^{(k)} = \text{Var}(\boldsymbol{\varepsilon}^{(k)}) = \mathcal{A}^{(k)} \text{Var}(\boldsymbol{\varepsilon}) \mathcal{A}^{(k)\top}.$$

In particular, if $\text{Var}(\varepsilon_t) = \sigma^2$ then $\Sigma^{(k)} = \mathcal{A}^{(k)}\mathcal{A}^{(k)\top}$. Furthermore, $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \Sigma)$ yields $\boldsymbol{\varepsilon}^{(k)} \sim \mathcal{N}(0, \Sigma^{(k)})$.

We can use

$$\mathbb{E}\|\tilde{S}^{(k)} - S^{(k)}\|^2 = \mathbb{E}\|\boldsymbol{\varepsilon}^{(k)}\|^2 = \text{tr } \Sigma^{(k)}.$$

Compute and plot $\text{tr } \Sigma^{(k)} = \text{tr}(\mathcal{A}^{(k)}\mathcal{A}^{(k)\top})$ for $k = 1, \dots, k_{\max}$.

Lemma 2.5. *It holds $(W^{(k)}W^{(k)\top})_{tt} \asymp (nh_k)^{-1}$.*

2.5.4 Regular design

Assume a [regular deterministic design](#) $t = 1/n, 2/n, \dots, 1$ and the vector $Y = (Y_t)$ is extended to $t \in [-0.5, 1.5]$; e.g. by $Y_t = Y_{1+t}$ for $t < 0$.

Lemma 2.6. *Assume a regular design with $N_k(t) \equiv N_k$ for $t \in [-0.5, 1.5]$. Then the operators $W^{(k)}$ from (2.1) and $\mathcal{A}^{(k)}$ from (2.2) mutually commute.*

Proof. Fix $k', k \geq 1$. Then for $t \in [0, 1]$

$$\begin{aligned} (W^{(k)} W^{(k')} Y)_t &= \sum_u \frac{K_{h_k}(t-u)}{N_k(t)} (W^{(k')} Y)_u \\ &= \sum_u \sum_s \frac{K_{h_k}(t-u)}{N_k(t)} \frac{K_{h_{k'}}(u-s)}{N_{k'}(u)} Y_s \\ &= \frac{1}{N_k N_{k'}} \sum_s \left\{ \sum_u K_{h_k}(t-u) K_{h_{k'}}(u-s) \right\} Y_s. \end{aligned}$$

The same representation applies to $(W^{(k')} W^{(k)} Y)_t$. The statement about $\mathcal{A}^{(k)}$ follows from (2.2). $\square$

3 Robust filtering

This section presents the robustified procedure based on local median filtering. More precisely, we apply a smoothed version of median filtering. Huber's proposal leads to

$$\rho(x) \stackrel{\text{def}}{=} \frac{1}{2} x^2 \mathbb{I}(|x| \leq H) + (Hx - \frac{1}{2} H^2) \mathbb{I}(|x| > H).$$

The value H is a tuning meta-parameter. Our theoretical results require 3 derivatives of ρ . We consider the third-order spline with $\rho(0) = \rho'(0) = 0$, $\rho''(0) = 1$, and

$$\rho'''(x) = -\text{sign}(x) \mathbb{I}(H \leq |x| \leq 2H).$$

The function $\rho(x)$ is quadratic for $|x| \leq H$ with $\rho(x) = x^2/2$, then the second derivative $\rho(x)$ decreases from one to zero at $x = 2H$. As $\rho''(x) \geq 0$, this function is convex.

Alternatively, with $\rho_0(x) = |x|$ and $\varphi_H(x) = \frac{1}{\sqrt{2\pi}H} \exp\left(-\frac{x^2}{2H^2}\right)$, one can use

$$\rho_H(x) \stackrel{\text{def}}{=} \rho_0 * \varphi_H(x).$$

This value can be computed analytically.

Lemma 3.1. *It holds*

$$\rho_H(x) = x \{2\Phi(x/H) - 1\} + \frac{2H}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2H^2}\right). \quad (3.1)$$

Also, for $x \geq 0$,

$$\begin{aligned} \rho'_H(x) &= 2\Phi(x/H) - 1, \\ \rho''_H(x) &= \frac{2}{H} \phi(x/H) = \frac{2}{H\sqrt{2\pi}} \exp\left(-\frac{x^2}{2H^2}\right), \\ \rho'''_H(x) &= -\frac{2x}{H^3\sqrt{2\pi}} \exp\left(-\frac{x^2}{2H^2}\right), \end{aligned}$$

Proof. We use

$$\rho_H(x) = \int_{-\infty}^{\infty} |t| \varphi_H(x-t) dt = H \int_{-\infty}^{\infty} |t| \varphi(x/H-t) dt = H \int_{-\infty}^{\infty} |t+x/H| \varphi(t) dt.$$

For any $a \geq 0$,

$$\begin{aligned} \int_{-\infty}^{\infty} |t+a| \varphi(t) dt &= \frac{1}{2} \int_{-\infty}^{\infty} (|t+a| + |t-a|) \varphi(t) dt = a \int_{-a}^a \varphi(t) dt + 2 \int_a^{\infty} t \varphi(t) dt \\ &= a \{ \Phi(a) - \Phi(-a) \} + \frac{2}{\sqrt{2\pi}} \int_a^{\infty} e^{-t^2/2} d(t^2/2) = a \{ \Phi(a) - \Phi(-a) \} + \frac{2}{\sqrt{2\pi}} e^{-a^2/2}. \end{aligned}$$

This implies (3.1). The other identities follow by direct differentiation. $\square$

Lemma 3.2. *Check that $\rho_H(x)$ is*

- *infinitely differentiable;*
- *symmetric, convex, positive, and monotonously increasing with $|x|$;*
- *it holds for all x*

$$\begin{aligned} |\rho'_H(x)| &\leq 1, \quad 1 - |\rho'_H(x)| \leq 2\{1 - \Phi(|x|/H)\}, \\ 0 \leq \rho''_H(x) &\leq \frac{2}{H\sqrt{2\pi}}; \quad |\rho'''_H(x)| \leq \frac{2}{H^2\sqrt{2\pi}}; \quad |\rho'''_H(x)| \leq \frac{|x|}{H^2} \rho''_H(x). \end{aligned}$$

- *for $|x| > 3H$, $|\rho_H(x) - |x||$ is very small:*

$$|\rho_H(x) - |x|| \leq \frac{2H}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2H^2}\right) \leq 0.009H, \quad |x| > 3H.$$

The procedure reads as follows:

- Set $Y^{(1)} \equiv Y$, $k = 1$;

- compute

$$\begin{aligned}\tilde{S}_t^{(k)} &= \operatorname{arginf}_x \sum_u \rho(Y_u^{(k)} - x) K_{h_k}(t - u); \\ Y_t^{(k+1)} &= Y_t^{(k)} - \tilde{S}_t^{(k)};\end{aligned}\tag{3.2}$$

- recompute $h_{k+1} = h_k/a$;
- increase k by 1; loop until $k > k_{\max}$ or $h_k < h_{\min}$;

Choice of H is important. This value should be of the same order as the standard deviation of the noise before contamination.

3.1 First-step decomposition

For a general ρ -function, one of the main questions is the definition of the true IMF. Define

$$S_t \stackrel{\text{def}}{=} \operatorname{arginf}_x \sum_u \mathbb{E} \rho(Y_u - x) K_h(t - u).$$

If $Y_t = X_t + \varepsilon_t$ for ε_t i.i.d. then

$$\mathbb{E} \rho(Y_u - x) = \phi(X_u - x), \quad \phi(x) \stackrel{\text{def}}{=} \mathbb{E} \rho(x + \varepsilon),$$

leading to

$$S_t = \operatorname{arginf}_x \sum_u \phi(X_u - x) K_h(t - u).$$

Lemma 3.3. ρ *convex* implies ϕ *convex*.

Define

$$N_t(x) \stackrel{\text{def}}{=} \sum_u \mathbb{E} \rho''(Y_u - x) K_h(t - u) = \sum_u \phi''(X_u - x) K_h(t - u) \tag{3.3}$$

and

$$N_t \stackrel{\text{def}}{=} N_t(X_t).$$

Also, denote

$$\diamond_{u,t} \stackrel{\text{def}}{=} \rho'(Y_u - X_t) - \mathbb{E} \rho'(Y_u - X_t) = \rho'(Y_u - X_t) - \phi'(X_u - X_t).$$

Obviously, $\mathbb{E}\diamond_{u,t} \equiv 0$. The following expansion will be a key tool of analysis:

$$\begin{aligned}\tilde{S}_t - S_t &\approx \frac{1}{N_t} \sum_u \diamond_{u,t} K_h(t-u) = \sum_u \diamond_{u,t} w_{u,t}, \\ w_{u,t} &\stackrel{\text{def}}{=} \frac{K_h(t-u)}{N_t}.\end{aligned}$$

3.2 IMFs expansions

Set $Y^{(1)} \equiv Y$ and $X^{(1)} \equiv X$.

Step $k \geq 1$: with $\varepsilon_u^{(k)} \stackrel{\text{def}}{=} Y_u^{(k)} - X_u^{(k)}$

$$\begin{aligned}S_t^{(k)} &= \operatorname{arginf}_x \sum_u \mathbb{E}\rho(Y_u^{(k)} - x) K_h(t-u) \\ &= \operatorname{arginf}_x \sum_u \mathbb{E}\rho(X_u^{(k)} + \varepsilon_u^{(k)} - x) K_h(t-u) \\ &= \operatorname{arginf}_x \sum_u \mathbb{E}\phi_k(X_u^{(k)} - x, u) K_h(t-u),\end{aligned}$$

where

$$\phi_k(x, u) \stackrel{\text{def}}{=} \mathbb{E}\rho(x + \varepsilon_u^{(k)}).$$

Continue with $Y^{(k+1)} = Y^{(k)} - \tilde{S}^{(k)}$ from (3.2) and

$$X^{(k+1)} = X^{(k)} - S^{(k)}.$$

4 Theoretical guarantees

This section describes the accuracy of the procedure. The starting point is an accuracy bounds for $\tilde{S}_t - S_t$ for a fixed $t \in [0, 1]$. We use tools from perturbed optimization; see Section A. Represent

$$\begin{aligned}\tilde{S}_t &= \operatorname{arginf}_x \sum_u \rho(Y_u - x) K_h(t-u) \\ S_t &= \operatorname{arginf}_x \sum_u \mathbb{E}\rho(Y_u - x) K_h(t-u) = \operatorname{arginf}_x \sum_u \phi(X_u - x) K_h(t-u).\end{aligned}\tag{4.1}$$

Also, define

$$\begin{aligned}\mathbb{F}_t &\stackrel{\text{def}}{=} \sum_u \phi''(X_u - S_t) K_h(t-u), \\ \nabla_t &\stackrel{\text{def}}{=} \sum_u \{\rho'(Y_u - S_t) - \mathbb{E}\rho'(Y_u - S_t)\} K_h(t-u),\end{aligned}$$

and $D_t \stackrel{\text{def}}{=} \mathbb{F}_t^{1/2}$. Note that $\mathbb{F} = N_t$; see (3.3).

Proposition 4.1. *On a random set $\Omega(\mathbf{x})$ with $\mathbb{P}(\Omega(\mathbf{x})) \geq 1 - e^{-x}$*

$$\sup_{t \in [0,1]} \|D_t(\tilde{S}_t - S_t + \mathbb{F}_t^{-1} \nabla_t)\| \leq , \quad .$$

4.1 Perturbation theory in action

Consider two optimization problems in (4.1) using the perturbed optimization approach.

We fix $t \in [0, 1]$. All the quantities considered below will depend on t . Define

$$\begin{aligned} f(x) &\stackrel{\text{def}}{=} \sum_u \phi(X_u - x) K_h(t - u), \\ f_\circ(x) &\stackrel{\text{def}}{=} \sum_u \rho(Y_u - x) K_h(t - u). \end{aligned} \tag{4.2}$$

The aim is to compare the solutions

$$x^* \stackrel{\text{def}}{=} \operatorname{arginf}_x f(x), \quad x_\circ \stackrel{\text{def}}{=} \operatorname{arginf}_x f_\circ(x).$$

Unfortunately, the perturbation $\zeta(x) \stackrel{\text{def}}{=} f(x) - f_\circ(x)$ is not linear in x . We apply Theorem A.9 to f and $f_\circ$ from (4.2). The functions $f(x)$ and $f_\circ(x)$ are strongly convex in x . Define

$$\begin{aligned} \mathbb{F}_\circ &\stackrel{\text{def}}{=} f_\circ''(x^*) = \sum_u \rho''(Y_u - x^*) K_h(t - u), \\ \mathbb{F} &\stackrel{\text{def}}{=} f''(x^*) = \sum_u \phi''(X_u - x^*) K_h(t - u). \end{aligned}$$

Also, define $D = \mathbb{F}^{1/2}$.

Lemma 4.2. *It holds on $\Omega(\mathbf{x})$*

(i)

$$\begin{aligned} |f'(x^*) - f'_\circ(x^*)| &= \frac{1}{D} \left| \sum_u \{ \rho'(Y_u - x^*) - \mathbb{E} \rho'(Y_u - x^*) \} K_h(t - u) \right| \leq \mathbf{C}_1, \\ |f''(x^*) - f''_\circ(x^*)| &= \frac{1}{D} \left| \sum_u \{ \rho''(Y_u - x^*) - \mathbb{E} \rho''(Y_u - x^*) \} K_h(t - u) \right| \leq \mathbf{C}_2, \end{aligned}$$

(ii)

$$\sup_{x \in U} \frac{1}{D^2} \left| \sum_u \rho'''(Y_u - x) K_h(t - u) \right| \leq \mathbf{C}_3,$$

(iii)

$$\sup_{x \in U} \frac{1}{D^2} \left| \sum_u \phi''(X_u - x) - \phi''(X_u - x^*) K_h(t - u) \right| \leq$$

Proof. This is a standard Bernstein-type deviation bound for independent sums under Lipschitz condition on each summand; see e.g. [Boucheron et al. \(2003\)](#). $\square$

Lemma 4.2 (ii) implies

$$\frac{1}{\mathbb{F}} |\mathbb{F}_\circ - \mathbb{F}| \leq \delta \stackrel{\text{def}}{=} \frac{\mathbb{C}_2}{D}. \quad (4.3)$$

Further, consider

$$\mathbf{b}_\circ \stackrel{\text{def}}{=} \frac{D}{\mathbb{F}_\circ} |f'_\circ(x^*)| = \frac{D}{\mathbb{F}_\circ} \left| \sum_u \rho'(Y_u - x^*) K_h(t - u) \right|.$$

As $f'(x^*) = 0$, it also holds

$$\mathbf{b}_\circ = \frac{D}{\mathbb{F}_\circ} \left| \sum_u \{ \rho'(Y_u - x^*) - \mathbb{E} \rho'(Y_u - x^*) \} K_h(t - u) \right|.$$

Lemma 4.2 (i) and (4.3) imply

$$\mathbf{b}_\circ \leq \frac{\mathbb{C}_1}{1 - \delta}.$$

Then, by (A.8) of Theorem A.9 with $\tau_3 = \mathbb{C}_3/D$,

$$|D^{-1} \mathbb{F}_\circ(x_\circ - x^* + \mathbb{F}_\circ^{-1} f'_\circ(x^*))| \leq \frac{3\mathbb{C}_3}{4D} \mathbf{b}_\circ^2 \quad (4.4)$$

yielding by (4.3)

$$|x_\circ - x^* + \mathbb{F}_\circ^{-1} f'_\circ(x^*)| \leq \frac{3\mathbb{C}_3}{4D^2(1 - \delta)} \mathbf{b}_\circ^2 \leq \frac{3\mathbb{C}_1^2 \mathbb{C}_3}{4D^2(1 - \delta)^3}.$$

Also,

$$|\mathbb{F}_\circ^{-1} f'_\circ(x^*) - \mathbb{F}^{-1} f'_\circ(x^*)| = \frac{|\mathbb{F}_\circ - \mathbb{F}|}{\mathbb{F}_\circ \mathbb{F}} |f'_\circ(x^*)| \leq \frac{\delta}{1 - \delta} \frac{1}{\mathbb{F}} |f'_\circ(x^*)| \leq \frac{\mathbb{C}_1 \mathbb{C}_2}{D^2(1 - \delta)}$$

Together with

Approximation (4.3) enables us to replace $\mathbb{F}_\circ$ with $\mathbb{F}$ everywhere and to rewrite the bound (4.4) as

$$|x_\circ - x^* + \mathbb{F}^{-1} f'_\circ(x^*)| \leq \frac{\mathbb{C}}{D^2}.$$

A Smooth perturbed optimization

This section presents conditions and accurate bounds for smooth perturbed optimization. We consider minimization of a smooth convex function f and study the impact of a linear, quadratic, or smooth perturbation.

A.1 Gateaux smoothness and self-concordance

Below we assume the function $f(\mathbf{v})$, $\mathbf{v} \in \mathcal{V} \subseteq \mathbb{R}^p$ to be strongly convex with a positive definite Hessian $\mathbb{F}(\mathbf{v}) \stackrel{\text{def}}{=} \nabla^2 f(\mathbf{v})$. Also, assume $f(\mathbf{v})$ three or sometimes even four times Gateaux differentiable in $\mathbf{v} \in \mathcal{V}$. Local smoothness of f will be described by the relative error of the Taylor expansion of the third or fourth order. Define

$$\begin{aligned}\delta_3(\mathbf{v}, \mathbf{u}) &= f(\mathbf{v} + \mathbf{u}) - f(\mathbf{v}) - \langle \nabla f(\mathbf{v}), \mathbf{u} \rangle - \frac{1}{2} \langle \nabla^2 f(\mathbf{v}), \mathbf{u}^{\otimes 2} \rangle, \\ \delta'_3(\mathbf{v}, \mathbf{u}) &= \langle \nabla f(\mathbf{v} + \mathbf{u}), \mathbf{u} \rangle - \langle \nabla f(\mathbf{v}), \mathbf{u} \rangle - \langle \nabla^2 f(\mathbf{v}), \mathbf{u}^{\otimes 2} \rangle,\end{aligned}$$

and

$$\delta_4(\mathbf{v}, \mathbf{u}) = f(\mathbf{v} + \mathbf{u}) - f(\mathbf{v}) - \langle \nabla f(\mathbf{v}), \mathbf{u} \rangle - \frac{1}{2} \langle \nabla^2 f(\mathbf{v}), \mathbf{u}^{\otimes 2} \rangle - \frac{1}{6} \langle \nabla^3 f(\mathbf{v}), \mathbf{u}^{\otimes 3} \rangle.$$

Now, for each $\mathbf{v}$, suppose to be given a positive symmetric matrix $D(\mathbf{v})$ defining a local metric and a local vicinity around $\mathbf{v}$: for some radius $\mathbf{r}$,

$$\mathcal{U}_{\mathbf{r}}(\mathbf{v}) = \{\mathbf{u} \in \mathbb{R}^p : \|D(\mathbf{v})\mathbf{u}\| \leq \mathbf{r}\}$$

Local smoothness properties of f at $\mathbf{v}$ are given via the quantities

$$\omega(\mathbf{v}) \stackrel{\text{def}}{=} \sup_{\mathbf{u} : \|D(\mathbf{v})\mathbf{u}\| \leq \mathbf{r}} \frac{2|\delta_3(\mathbf{v}, \mathbf{u})|}{\|D(\mathbf{v})\mathbf{u}\|^2}, \quad \omega'(\mathbf{v}) \stackrel{\text{def}}{=} \sup_{\mathbf{u} : \|D(\mathbf{v})\mathbf{u}\| \leq \mathbf{r}} \frac{|\delta'_3(\mathbf{v}, \mathbf{u})|}{\|D(\mathbf{v})\mathbf{u}\|^2}. \quad (\text{A.1})$$

The definition yields for any $\mathbf{u}$ with $\|D(\mathbf{v})\mathbf{u}\| \leq \mathbf{r}$

$$|\delta_3(\mathbf{v}, \mathbf{u})| \leq \frac{\omega(\mathbf{v})}{2} \|D(\mathbf{v})\mathbf{u}\|^2, \quad |\delta'_3(\mathbf{v}, \mathbf{u})| \leq \omega'(\mathbf{v}) \|D(\mathbf{v})\mathbf{u}\|^2. \quad (\text{A.2})$$

The approximation results can be improved under a third-order upper bound on the error of Taylor expansion.

($\mathcal{T}_3$) For some τ_3

$$|\delta_3(\mathbf{v}, \mathbf{u})| \leq \frac{\tau_3}{6} \|D(\mathbf{v})\mathbf{u}\|^3, \quad |\delta'_3(\mathbf{v}, \mathbf{u})| \leq \frac{\tau_3}{2} \|D(\mathbf{v})\mathbf{u}\|^3, \quad \mathbf{u} \in \mathcal{U}_{\mathbf{r}}(\mathbf{v}).$$

($\mathcal{T}_4$) For some τ_4

$$|\delta_4(\mathbf{v}, \mathbf{u})| \leq \frac{\tau_4}{24} \|D(\mathbf{v}) \mathbf{u}\|^4, \quad \mathbf{u} \in \mathcal{U}_{\mathbf{r}}(\mathbf{v}).$$

We also present a version of ($\mathcal{T}_3$) resp. ($\mathcal{T}_4$) in terms of the third (resp. fourth) derivative of f .

($\mathcal{T}_3^*$) $f(\mathbf{v})$ is three times differentiable and

$$\sup_{\mathbf{u}: \|D(\mathbf{v})\mathbf{u}\| \leq \mathbf{r}} \sup_{\mathbf{z} \in \mathbb{R}^p} \frac{|\langle \nabla^3 f(\mathbf{v} + \mathbf{u}), \mathbf{z}^{\otimes 3} \rangle|}{\|D(\mathbf{v})\mathbf{z}\|^3} \leq \tau_3.$$

($\mathcal{T}_4^*$) $f(\mathbf{v})$ is four times differentiable and

$$\sup_{\mathbf{u}: \|D(\mathbf{v})\mathbf{u}\| \leq \mathbf{r}} \sup_{\mathbf{z} \in \mathbb{R}^p} \frac{|\langle \nabla^4 f(\mathbf{v} + \mathbf{u}), \mathbf{z}^{\otimes 4} \rangle|}{\|D(\mathbf{v})\mathbf{z}\|^4} \leq \tau_4.$$

By Banach's characterization [Banach \(1938\)](#), ($\mathcal{T}_3^*$) implies

$$|\langle \nabla^3 f(\mathbf{v} + \mathbf{u}), \mathbf{z}_1 \otimes \mathbf{z}_2 \otimes \mathbf{z}_3 \rangle| \leq \tau_3 \|D(\mathbf{v})\mathbf{z}_1\| \|D(\mathbf{v})\mathbf{z}_2\| \|D(\mathbf{v})\mathbf{z}_3\|$$

for any $\mathbf{u}$ with $\|D(\mathbf{v})\mathbf{u}\| \leq \mathbf{r}$ and all $\mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3 \in \mathbb{R}^p$. Similarly under ($\mathcal{T}_4^*$)

$$\begin{aligned} & |\langle \nabla^4 f(\mathbf{v} + \mathbf{u}), \mathbf{z}_1 \otimes \mathbf{z}_2 \otimes \mathbf{z}_3 \otimes \mathbf{z}_4 \rangle| \\ & \leq \tau_4 \|D(\mathbf{v})\mathbf{z}_1\| \|D(\mathbf{v})\mathbf{z}_2\| \|D(\mathbf{v})\mathbf{z}_3\| \|D(\mathbf{v})\mathbf{z}_4\|, \quad \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3, \mathbf{z}_4 \in \mathbb{R}^p. \end{aligned}$$

Lemma A.1. Under ($\mathcal{T}_3$) or ($\mathcal{T}_3^*$), it holds for $\omega(\mathbf{v})$ and $\omega'(\mathbf{v})$ from (A.1)

$$\omega(\mathbf{v}) \leq \frac{\tau_3 \mathbf{r}}{3}, \quad \omega'(\mathbf{v}) \leq \frac{\tau_3 \mathbf{r}}{2}.$$

Proof. For any $\mathbf{u} \in \mathcal{U}_{\mathbf{r}}(\mathbf{v})$ with $\|D(\mathbf{v})\mathbf{u}\| \leq \mathbf{r}$

$$|\delta_3(\mathbf{v}, \mathbf{u})| \leq \frac{\tau_3}{6} \|D(\mathbf{v})\mathbf{u}\|^3 \leq \frac{\tau_3 \mathbf{r}}{6} \|D(\mathbf{v})\mathbf{u}\|^2,$$

and the bound for $\omega(\mathbf{v})$ follows. The proof for $\omega'(\mathbf{v})$ is similar. Under ($\mathcal{T}_3^*$), apply the third order Taylor expansion to the univariate function $f(\mathbf{v} + t\mathbf{u})$ of t and ($\mathcal{T}_3^*$) with $\mathbf{z} \equiv \mathbf{u}$. $\square$

The values τ_3 and τ_4 are usually very small. Some quantitative bounds are given later in this section under the assumption that the function $f(\mathbf{v})$ can be written in the form $f(\mathbf{v}) = nh(\mathbf{v})$ for a fixed smooth function $h(\mathbf{v})$ with the Hessian $\nabla^2 h(\mathbf{v})$. The factor n has meaning of the sample size; see Section ?? or Section ??.

$(\mathcal{S}_3^*)$ $f(\mathbf{v}) = nh(\mathbf{v})$ for $h(\mathbf{v})$ three times differentiable and for a metric tensor $\mathbf{g}(\mathbf{v})$

$$\sup_{\mathbf{u}: \|\mathbf{g}(\mathbf{v})\mathbf{u}\| \leq \mathbf{r}/\sqrt{n}} \frac{|\langle \nabla^3 h(\mathbf{v} + \mathbf{u}), \mathbf{u}^{\otimes 3} \rangle|}{\|\mathbf{g}(\mathbf{v})\mathbf{u}\|^3} \leq \mathbf{c}_3.$$

$(\mathcal{S}_4^*)$ the function $h(\cdot)$ satisfies $(\mathcal{S}_3^*)$ and

$$\sup_{\mathbf{u}: \|\mathbf{g}(\mathbf{v})\mathbf{u}\| \leq \mathbf{r}/\sqrt{n}} \frac{|\langle \nabla^4 h(\mathbf{v} + \mathbf{u}), \mathbf{u}^{\otimes 4} \rangle|}{\|\mathbf{g}(\mathbf{v})\mathbf{u}\|^4} \leq \mathbf{c}_4.$$

$(\mathcal{S}_3^*)$ and $(\mathcal{S}_4^*)$ are local versions of the so-called self-concordance condition; see [Nesterov and Nemirovskii \(1994\)](#) and [Ostrovskii and Bach \(2021\)](#). In fact, they require that each univariate function $h(\mathbf{v} + t\mathbf{u})$ of $t \in \mathbb{R}$ is self-concordant with some universal constants $\mathbf{c}_3$ and $\mathbf{c}_4$. Under $(\mathcal{S}_3^*)$ and $(\mathcal{S}_4^*)$, with $D^2(\mathbf{v}) = n\mathbf{g}^2(\mathbf{v})$, the values $\delta_3(\mathbf{v}, \mathbf{u})$, $\delta_4(\mathbf{v}, \mathbf{u})$, and $\omega(\mathbf{v})$, $\omega'(\mathbf{v})$ can be well bounded.

Lemma A.2. *Suppose $(\mathcal{S}_3^*)$. Then $(\mathcal{T}_3)$ follows with $\tau_3 = \mathbf{c}_3 n^{-1/2}$. Moreover, for $\omega(\mathbf{v})$ and $\omega'(\mathbf{v})$ from (A.1), it holds*

$$\omega(\mathbf{v}) \leq \frac{\mathbf{c}_3 \mathbf{r}}{3n^{1/2}}, \quad \omega'(\mathbf{v}) \leq \frac{\mathbf{c}_3 \mathbf{r}}{2n^{1/2}}. \quad (\text{A.3})$$

Also $(\mathcal{T}_4)$ follows from $(\mathcal{S}_4^*)$ with $\tau_4 = \mathbf{c}_4 n^{-1}$.

Proof. For any $\mathbf{u} \in \mathcal{U}_{\mathbf{r}}(\mathbf{v})$, by third order Taylor expansion,

$$\begin{aligned} |\delta_3(\mathbf{v}, \mathbf{u})| &\leq \sup_{t \in [0,1]} \frac{1}{6} |\langle \nabla^3 f(\mathbf{v} + t\mathbf{u}), \mathbf{u}^{\otimes 3} \rangle| = \frac{n}{6} \sup_{t \in [0,1]} |\langle \nabla^3 h(\mathbf{v} + t\mathbf{u}), \mathbf{u}^{\otimes 3} \rangle| \\ &\leq \frac{n \mathbf{c}_3}{6} \|\mathbf{g}(\mathbf{v})\mathbf{u}\|^3 = \frac{n^{-1/2} \mathbf{c}_3}{6} \|D(\mathbf{v})\mathbf{u}\|^3 \leq \frac{n^{-1/2} \mathbf{c}_3 \mathbf{r}}{6} \|D(\mathbf{v})\mathbf{u}\|^2. \end{aligned}$$

This implies $(\mathcal{T}_3)$ as well as (A.3); see (A.2). The statement about $(\mathcal{T}_4)$ is similar. $\square$

A.2 Perturbed optimization

This section provides an overview of some results from [Spokoiny \(2025\)](#). Let $f(\mathbf{v})$ be a smooth strongly convex function,

$$\mathbf{v}^* = \underset{\mathbf{v}}{\operatorname{arginf}} f(\mathbf{v}),$$

and $\mathbb{F} = \nabla^2 f(\mathbf{v}^*)$. We study the question of how the point of minimum and the value of minimum of f change under some perturbation of f . More precisely, with $f^\circ(\mathbf{v})$

being a perturbed version of $f(\mathbf{v})$, we consider in parallel two optimization problems

$$\mathbf{v}^* = \underset{\mathbf{v}}{\operatorname{arginf}} f(\mathbf{v}), \quad \mathbf{v}^\circ = \underset{\mathbf{v}}{\operatorname{arginf}} f^\circ(\mathbf{v}).$$

The goal is to describe the discrepancies $\mathbf{v}^\circ - \mathbf{v}^*$ and $f^\circ(\mathbf{v}^\circ) - f^\circ(\mathbf{v}^*)$ induced by this perturbation. We also present the results for two specific cases: a linear and a quadratic perturbation.

A.3 Smoothness conditions

Assume $f(\mathbf{v})$ to be convex and three or sometimes even four times Gateaux differentiable in $\mathbf{v} \in \mathcal{Y}$. For a fixed $\mathbf{v} \in \mathcal{Y}$, we need such a condition uniformly in a local vicinity of a fixed point $\mathbf{v} \in \mathcal{Y}$. The notion of locality is given in terms of a metric tensor $D(\mathbf{v}) \in \mathfrak{M}_p$, where $\mathfrak{M}_p$ is the set of symmetric positive p -matrices. In most of the results later on, one can use $D(\mathbf{v}) = \mathbb{F}^{1/2}(\mathbf{v})$, where $\mathbb{F}(\mathbf{v}) = \nabla^2 f(\mathbf{v})$. Introduce the following conditions.

$(\mathcal{T}_3^*)$ $f(\mathbf{v} + \mathbf{u})$ is three times differentiable in $\mathbf{u}$ and with D possibly depending on $\mathbf{v}$

$$\sup_{\mathbf{u}: \|D\mathbf{u}\| \leq \mathbf{r}} \sup_{\mathbf{z} \in \mathbb{R}^p} \frac{|\langle \nabla^3 f(\mathbf{v} + \mathbf{u}), \mathbf{z}^{\otimes 3} \rangle|}{\|D\mathbf{z}\|^3} \leq \tau_3.$$

$(\mathcal{T}_4^*)$ $f(\mathbf{v} + \mathbf{u})$ is four times differentiable in $\mathbf{u}$ and

$$\sup_{\mathbf{u}: \|D\mathbf{u}\| \leq \mathbf{r}} \sup_{\mathbf{z} \in \mathbb{R}^p} \frac{|\langle \nabla^4 f(\mathbf{v} + \mathbf{u}), \mathbf{z}^{\otimes 4} \rangle|}{\|D\mathbf{z}\|^4} \leq \tau_4.$$

By Banach's characterization [Banach \(1938\)](#), $(\mathcal{T}_3^*)$ implies

$$|\langle \nabla^3 f(\mathbf{v} + \mathbf{u}), \mathbf{z}_1 \otimes \mathbf{z}_2 \otimes \mathbf{z}_3 \rangle| \leq \tau_3 \|D\mathbf{z}_1\| \|D\mathbf{z}_2\| \|D\mathbf{z}_3\|$$

for any $\mathbf{u}$ with $\|D\mathbf{u}\| \leq \mathbf{r}$ and all $\mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3 \in \mathbb{R}^p$.

The value τ_3 is usually very small. If the function $f(\mathbf{v})$ can be written in the form $f(\mathbf{v}) = nh(\mathbf{v})$ for a fixed smooth function $h(\mathbf{v})$ with the Hessian $\nabla^2 h(\mathbf{v})$, and a factor n meaning of the sample size, then $\tau_3 \asymp n^{-1/2}$. $(\mathcal{T}_3^*)$ is a local version of the so-called self-concordance condition; see [Nesterov \(1988\)](#) and [Ostrovskii and Bach \(2021\)](#).

A.4 A linear perturbation

For the objective function f , let another function $f^\circ(\mathbf{v})$ satisfy for some vector $\mathbf{A}$

$$f^\circ(\mathbf{v}) - f^\circ(\mathbf{v}^*) = \langle \mathbf{v} - \mathbf{v}^*, \mathbf{A} \rangle + f(\mathbf{v}) - f(\mathbf{v}^*). \quad (\text{A.4})$$

Define

$$\mathbf{v}^\circ \stackrel{\text{def}}{=} \underset{\mathbf{v}}{\operatorname{arginf}} f^\circ(\mathbf{v}), \quad f^\circ(\mathbf{v}^\circ) = \inf_{\mathbf{v}} f^\circ(\mathbf{v}). \quad (\text{A.5})$$

The analysis aims to evaluate the quantities $\mathbf{v}^\circ - \mathbf{v}^*$ and $f^\circ(\mathbf{v}^\circ) - f^\circ(\mathbf{v}^*)$.

A.4.1 Local concentration

Let f satisfy (A.1) at $\mathbf{v}^*$ with $D(\mathbf{v}^*) = D \leq \varkappa \mathbb{F}^{1/2}$ for some $\varkappa > 0$. The latter means that the matrix $\mathbb{F} - \varkappa^2 D^2$ is positive definite. The next result describes the concentration properties of $\mathbf{v}^\circ$ from (A.5) in a local elliptic set

$$\mathcal{A}(\mathbf{r}) \stackrel{\text{def}}{=} \{\mathbf{v} : \|\mathbb{F}^{1/2}(\mathbf{v} - \mathbf{v}^*)\| \leq \mathbf{r}\},$$

where $\mathbf{r}$ is slightly larger than $\|\mathbb{F}^{-1/2}\mathbf{A}\|$.

Theorem A.3. *Let $f(\mathbf{v})$ be a strongly convex function with $f(\mathbf{v}^*) = \inf_{\mathbf{v}} f(\mathbf{v})$ and $\mathbb{F} = \nabla^2 f(\mathbf{v}^*)$. Let, further, $f^\circ(\mathbf{v})$ and $f(\mathbf{v})$ be related by (A.4) with some vector $\mathbf{A}$. Fix $\nu < 1$ and $\mathbf{r}$ such that $\|\mathbb{F}^{-1/2}\mathbf{A}\| \leq \nu \mathbf{r}$. Suppose now that $f(\mathbf{v})$ satisfy (A.1) for $\mathbf{v} = \mathbf{v}^*$, $D(\mathbf{v}^*) = D \leq \varkappa \mathbb{F}^{1/2}$ with some $\varkappa > 0$ and ω' such that*

$$1 - \nu - \omega' \varkappa^2 > 0.$$

Then for $\mathbf{v}^\circ$ from (A.5), it holds

$$\|\mathbb{F}^{1/2}(\mathbf{v}^\circ - \mathbf{v}^*)\| \leq \mathbf{r} \quad \text{and} \quad \|D(\mathbf{v}^\circ - \mathbf{v}^*)\| \leq \varkappa \mathbf{r}.$$

A.4.2 Second-order expansions

The result of Theorem A.3 allows to localize the point $\mathbf{v}^\circ = \underset{\mathbf{v}}{\operatorname{arginf}} f^\circ(\mathbf{v})$ in the local vicinity $\mathcal{A}(\mathbf{r})$ of $\mathbf{v}^*$. The use of smoothness properties of f° or, equivalently, of f , in this vicinity helps to obtain rather sharp expansions for $\mathbf{v}^\circ - \mathbf{v}^*$ and for $f^\circ(\mathbf{v}^\circ) - f^\circ(\mathbf{v}^*)$.

Theorem A.4. *Assume the conditions of Theorem A.3 with $\omega \leq 1/3$. Then*

$$\begin{aligned} -\frac{\omega}{1 - \varkappa^2 \omega} \|D \mathbb{F}^{-1} \mathbf{A}\|^2 &\leq 2f^\circ(\mathbf{v}^\circ) - 2f^\circ(\mathbf{v}^*) + \|\mathbb{F}^{-1/2} \mathbf{A}\|^2 \\ &\leq \frac{\omega}{1 + \varkappa^2 \omega} \|D \mathbb{F}^{-1} \mathbf{A}\|^2. \end{aligned}$$

Also

$$\begin{aligned} \|D(\mathbf{v}^\circ - \mathbf{v}^* + \mathbb{F}^{-1} \mathbf{A})\| &\leq \frac{2\sqrt{\omega}}{1 - \varkappa^2 \omega} \|D \mathbb{F}^{-1} \mathbf{A}\|, \\ \|D(\mathbf{v}^\circ - \mathbf{v}^*)\| &\leq \frac{1 + 2\sqrt{\omega}}{1 - \varkappa^2 \omega} \|D \mathbb{F}^{-1} \mathbf{A}\|. \end{aligned}$$

A.4.3 Expansions under third order smoothness

The results of Theorem A.4 can be refined if f satisfies condition $(\mathcal{T}_3)$.

Theorem A.5. *Let $f(\mathbf{v})$ be a strongly convex function with $f(\mathbf{v}^*) = \inf_{\mathbf{v}} f(\mathbf{v})$ and $\mathbb{F} = \nabla^2 f(\mathbf{v}^*)$. Let $f^\circ(\mathbf{v})$ fulfill (A.4) with some vector $\mathbf{A}$. Suppose that $f(\mathbf{v})$ follows $(\mathcal{T}_3)$ at $\mathbf{v}^*$ with D^2 , $\mathbf{r}$, and τ_3 such that*

$$D^2 \leq \kappa^2 \mathbb{F}, \quad \mathbf{r} \geq \frac{4\kappa}{3} \|\mathbb{F}^{-1/2} \mathbf{A}\|, \quad \kappa^3 \tau_3 \|\mathbb{F}^{-1/2} \mathbf{A}\| < \frac{1}{4}.$$

Then $\mathbf{v}^\circ = \operatorname{arginf}_{\mathbf{v}} f^\circ(\mathbf{v})$ satisfies

$$\|\mathbb{F}^{1/2}(\mathbf{v}^\circ - \mathbf{v}^*)\| \leq \frac{4}{3} \|\mathbb{F}^{-1/2} \mathbf{A}\|, \quad \|D(\mathbf{v}^\circ - \mathbf{v}^*)\| \leq \frac{4\kappa}{3} \|\mathbb{F}^{-1/2} \mathbf{A}\|.$$

Moreover,

$$\left| 2f^\circ(\mathbf{v}^\circ) - 2f^\circ(\mathbf{v}^*) + \|\mathbb{F}^{-1/2} \mathbf{A}\|^2 \right| \leq \frac{\tau_3}{2} \|D \mathbb{F}^{-1} \mathbf{A}\|^3.$$

A.4.4 Advanced approximation under locally uniform smoothness

The bounds of Theorem A.5 can be made more accurate if f follows $(\mathcal{T}_3^*)$ and $(\mathcal{T}_4^*)$ and one can apply Taylor's expansion around any point close to $\mathbf{v}^*$. In particular, the improved results do not involve the value $\|\mathbb{F}^{-1/2} \mathbf{A}\|$ which can be large or even infinite in some situation; see Section A.5.

Theorem A.6. *Let $f(\mathbf{v})$ be a strongly convex function with $f(\mathbf{v}^*) = \min_{\mathbf{v}} f(\mathbf{v})$ and $\mathbb{F} = \nabla^2 f(\mathbf{v}^*)$. Assume $(\mathcal{T}_3^*)$ at $\mathbf{v}^*$ with D^2 , $\mathbf{r}$, and τ_3 such that*

$$D^2 \leq \kappa^2 \mathbb{F}, \quad \mathbf{r} \geq \frac{3}{2} \|D \mathbb{F}^{-1} \mathbf{A}\|, \quad \kappa^2 \tau_3 \|D \mathbb{F}^{-1} \mathbf{A}\| < \frac{4}{9}.$$

Then

$$\begin{aligned} \|D(\mathbf{v}^\circ - \mathbf{v}^*)\| &\leq \frac{3}{2} \|D \mathbb{F}^{-1} \mathbf{A}\|, \\ \|D^{-1} \mathbb{F}(\mathbf{v}^\circ - \mathbf{v}^* + \mathbb{F}^{-1} \mathbf{A})\| &\leq \frac{3\tau_3}{4} \|D \mathbb{F}^{-1} \mathbf{A}\|^2. \end{aligned} \tag{A.6}$$

Moreover,

$$\left| 2f^\circ(\mathbf{v}^\circ) - 2f^\circ(\mathbf{v}^*) + \|\mathbb{F}^{-1/2} \mathbf{A}\|^2 \right| \leq \frac{\tau_3}{2} \|D \mathbb{F}^{-1} \mathbf{A}\|^3.$$

Remark A.1. The roles of f and f° can be exchanged. In particular, (A.6) applies with $\mathbb{F} = \mathbb{F}(\mathbf{v}^\circ)$ provided that $(\mathcal{T}_3^*)$ is also fulfilled at $\mathbf{v}^\circ$.

A.5 Quadratic penalization

Here we discuss the case when $f^\circ(\mathbf{v}) - f(\mathbf{v})$ is quadratic. The general case can be reduced to the situation with $f^\circ(\mathbf{v}) = f(\mathbf{v}) - \|G\mathbf{v}\|^2/2$. To make the dependence of G more explicit, denote $f_G(\mathbf{v}) \stackrel{\text{def}}{=} f(\mathbf{v}) - \|G\mathbf{v}\|^2/2$,

$$\mathbf{v}^* = \underset{\mathbf{v}}{\operatorname{arginf}} f(\mathbf{v}), \quad \mathbf{v}_G^* = \underset{\mathbf{v}}{\operatorname{arginf}} f_G(\mathbf{v}) = \underset{\mathbf{v}}{\operatorname{arginf}} \{f(\mathbf{v}) - \|G\mathbf{v}\|^2/2\}.$$

We study the bias $\mathbf{v}_G^* - \mathbf{v}^*$ induced by this penalization. Now we turn to the general case with f satisfying $(\mathcal{T}_3^*)$. Define

$$\mathbb{F}_G \stackrel{\text{def}}{=} \nabla^2 f_G(\mathbf{v}^*), \quad \mathbf{b}_G \stackrel{\text{def}}{=} \|D \mathbb{F}_G^{-1} G^2 \mathbf{v}^*\|.$$

Proposition A.7. *Let $f_G(\mathbf{v}) = f(\mathbf{v}) - \|G\mathbf{v}\|^2/2$ be convex and follow $(\mathcal{T}_3^*)$ with some D^2 , τ_3 , and $\mathbf{r}$ satisfying for $\kappa > 0$*

$$D^2 \leq \kappa^2 \mathbb{F}_G, \quad \mathbf{r} \geq 3\mathbf{b}_G/2, \quad \kappa^2 \tau_3 \mathbf{b}_G < 4/9,$$

where $\mathbf{b}_G \stackrel{\text{def}}{=} \|D \mathbb{F}_G^{-1} G^2 \mathbf{v}^*\|$. Then

$$\begin{aligned} \|D(\mathbf{v}_G^* - \mathbf{v}^*)\| &\leq \frac{3}{2} \mathbf{b}_G, \\ \|D^{-1} \mathbb{F}_G(\mathbf{v}_G^* - \mathbf{v}^* + \mathbb{F}_G^{-1} G^2 \mathbf{v}^*)\| &\leq \frac{3\tau_3}{4} \mathbf{b}_G^2, \\ \left| 2f_G(\mathbf{v}_G^*) - 2f_G(\mathbf{v}^*) - \frac{1}{2} \|\mathbb{F}_G^{-1/2} G^2 \mathbf{v}^*\|^2 \right| &\leq \frac{\tau_3}{2} \mathbf{b}_G^3. \end{aligned} \tag{A.7}$$

A.6 A smooth penalty

The case of a general smooth penalty $\operatorname{pen}_G(\mathbf{v})$ can be studied similarly to the quadratic case. Denote $f_G(\mathbf{v}) \stackrel{\text{def}}{=} f(\mathbf{v}) + \operatorname{pen}_G(\mathbf{v})$,

$$\mathbf{v}^* = \underset{\mathbf{v}}{\operatorname{arginf}} f(\mathbf{v}), \quad \mathbf{v}_G^* = \underset{\mathbf{v}}{\operatorname{arginf}} f_G(\mathbf{v}) = \underset{\mathbf{v}}{\operatorname{arginf}} \{f(\mathbf{v}) + \operatorname{pen}_G(\mathbf{v})\}.$$

We study the bias $\mathbf{v}_G^* - \mathbf{v}^*$ induced by this penalization. The statement of Theorem A.7 and its proof can be easily extended to this situation. Define

$$\mathbb{F}_G \stackrel{\text{def}}{=} \nabla^2 f_G(\mathbf{v}^*), \quad \mathbf{b}_G \stackrel{\text{def}}{=} \|D \mathbb{F}_G^{-1} \mathbf{M}_G\|, \quad \mathbf{M}_G \stackrel{\text{def}}{=} \nabla \operatorname{pen}_G(\mathbf{v}^*).$$

Theorem A.8. *Let $f_G(\mathbf{v}) = f(\mathbf{v}) + \operatorname{pen}_G(\mathbf{v})$ be strongly convex and follow $(\mathcal{T}_3^*)$ at $\mathbf{v}^*$ with some D^2 , τ_3 , and $\mathbf{r}$ satisfying for $\kappa > 0$*

$$D^2 \leq \kappa^2 \mathbb{F}_G, \quad \mathbf{r} \geq 3\mathbf{b}_G/2, \quad \kappa^2 \tau_3 \mathbf{b}_G < 4/9,$$

Then

$$\|D(\mathbf{v}_G^* - \mathbf{v}^*)\| \leq 3\mathbf{b}_G/2.$$

Moreover,

$$\begin{aligned} \|D^{-1}\mathbb{F}_G(\mathbf{v}_G^* - \mathbf{v}^* + \mathbb{F}_G^{-1}\mathbf{M}_G)\| &\leq \frac{3\tau_3}{4} \mathbf{b}_G^2, \\ \left| 2f_G(\mathbf{v}_G^*) - 2f_G(\mathbf{v}^*) + \|\mathbb{F}_G^{-1/2}\mathbf{M}_G\|^2 \right| &\leq \frac{\tau_3}{2} \mathbf{b}_G^3. \end{aligned}$$

Proof. Consider $f_G^\circ(\mathbf{v}) = f_G(\mathbf{v}) - \langle \nabla \text{pen}_G(\mathbf{v}^*), \mathbf{v} \rangle$. Then f_G° is strongly convex and $\nabla f_G^\circ(\mathbf{v}^*) = 0$ yielding $\mathbf{v}^* = \arg\inf_{\mathbf{v}} f_G^\circ(\mathbf{v})$. Also, $f_G(\mathbf{v})$ is a linear perturbation of $f_G^\circ(\mathbf{v})$ with $\mathbf{A} = \mathbf{M}_G = \nabla \text{pen}_G(\mathbf{v}^*)$. Now all the statements of Theorem A.7 apply to $\mathbf{v}_G^*$ with obvious changes. $\square$

A.7 A generic perturbation

This section presents a generic perturbation lemma for (A.5) without any assumption on the structure of the perturbation $f^\circ(\mathbf{v}) - f(\mathbf{v})$. Define

$$\mathbb{F}_\circ \stackrel{\text{def}}{=} \nabla^2 f^\circ(\mathbf{v}^*), \quad \mathbf{b}_\circ \stackrel{\text{def}}{=} \|D \mathbb{F}_\circ^{-1} \nabla f^\circ(\mathbf{v}^*)\|.$$

Theorem A.9. *Let a function $f^\circ(\mathbf{v})$ be strongly convex and follow $(\mathcal{T}_3^*)$ at $\mathbf{v}^*$ with some D^2 , τ_3 , and $\mathbf{r}$ satisfying for $\varkappa > 0$*

$$D^2 \leq \varkappa^2 \mathbb{F}_\circ, \quad \mathbf{r} \geq 3\mathbf{b}_\circ/2, \quad \varkappa^2 \tau_3 \mathbf{b}_\circ < 4/9.$$

Then

$$\begin{aligned} \|D(\mathbf{v}^\circ - \mathbf{v}^*)\| &\leq \frac{3}{2} \mathbf{b}_\circ, \\ \|D^{-1}\mathbb{F}_\circ(\mathbf{v}^\circ - \mathbf{v}^* + \mathbb{F}_\circ^{-1}\nabla f^\circ(\mathbf{v}^*))\| &\leq \frac{3\tau_3}{4} \mathbf{b}_\circ^2, \\ \left| 2f^\circ(\mathbf{v}^\circ) - 2f^\circ(\mathbf{v}^*) + \|\mathbb{F}_\circ^{-1/2}\nabla f^\circ(\mathbf{v}^*)\|^2 \right| &\leq \frac{\tau_3}{2} \mathbf{b}_\circ^3. \end{aligned} \tag{A.8}$$

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